

Supporting Information

Rapid sequential touches complicate whisking-phase analysis in mouse active touch

Yuxuan Xu

Additional results

Supplemental Table S1. Dense-to-isolated touch profile

Every indexed touch in this profile had an observed preceding onset and an observed following onset in the same trial. The same minimum interval was applied on both sides. Values summarize the mean of archive-label median file means for spike counts during 0-50 ms after touch. Uncertainty is reported as a bootstrap 95% confidence interval (CI).

Minimum interval to both neighbors (ms)	Touches	Files	Archive subject labels	Mean spikes per touch	Bootstrap 95% CI
0	16,231	41	21	0.364	0.241 to 0.502
20	15,750	41	21	0.362	0.239 to 0.497
30	14,808	41	21	0.360	0.240 to 0.491
50	10,097	41	21	0.363	0.247 to 0.490
100	4,532	41	21	0.367	0.247 to 0.501
250	350	38	21	0.350	0.214 to 0.506

Supplemental Table S2. Neighbor-threshold sensitivity of phase prediction

Neighbor-onset exclusion threshold (ms)	Retained touches	Archive labels, context model	Median phase gain, context	Median phase gain, kinematic adjustment
0	19,675	21	0.0124	0.0096
20	19,160	21	0.0124	0.0082
30	18,157	21	0.0134	0.0105
50	12,773	20	0.0152	0.0084
100	6,251	17	0.0086	0.0083
250	1,014	0	not estimated	not estimated

At 250 ms, no recording file met the requirements of at least 80 observations and at least 15 observations in each phase branch. The descriptive response remains available in Supplemental Table S1.

The reference phase analysis treats a trial-boundary side without an observed neighboring onset as having no recorded close neighbor. At the 50-ms threshold, a stricter sensitivity analysis required an observed onset on each side. It selected 10,097 touches; 9,247 touches from 30 files and 19 archive subject labels met the phase-model criteria. Median phase gains were 0.0203 after contextual adjustment (95% CI, 0.0046 to 0.0336; $P < 0.001$) and 0.0079 after adjustment for measured movement (95% CI, 0.0008 to 0.0212; $P = 0.002$).

Supplemental Table S3. Baseline and response-window sensitivity

The table crosses all response and baseline windows. One condition retained all preceding intervals; the comparison condition required a preceding interval of at least 100 ms. For each response width, touches with a following onset inside that window were excluded. Coefficients are standardized coefficients for the log preceding inter-touch interval. Q values control the false discovery rate across these 50 sensitivity estimates. The reference $q = 0.017$ in the manuscript comes from a 30-cell grid spanning six interval thresholds and five baseline widths. Five cells at the 250-ms threshold contained fewer than five archive labels, so 25 comparisons entered that adjustment. The minimum of five labels avoids reporting group tests from fewer than five archive grouping keys; smaller cells remain descriptive. The $q = 0.010$ reported here comes from the separate 50-estimate response-by-baseline sensitivity family. Complete numerical results, sample counts, and confidence intervals are included with the numerical source data.

Response (ms)	Baseline (ms)	No-minimum coefficient	No-minimum q	100-ms coefficient	100-ms q
10	10	0.037	0.035	0.014	0.844
10	20	0.077	0.010	0.010	0.751
10	30	0.085	0.009	0.012	0.751
10	40	0.084	0.009	0.012	0.751
10	50	0.080	0.009	0.015	0.603
20	10	0.088	0.010	0.053	0.350
20	20	0.098	0.009	0.053	0.228
20	30	0.110	0.009	0.030	0.291
20	40	0.120	0.009	0.020	0.279
20	50	0.117	0.009	0.046	0.200
30	10	0.097	0.011	0.027	0.751
30	20	0.108	0.010	0.011	0.751
30	30	0.125	0.009	0.008	0.822
30	40	0.121	0.009	0.002	0.797
30	50	0.116	0.010	0.016	0.751

Response (ms)	Baseline (ms)	No-minimum coefficient	No-minimum q	100-ms coefficient	100-ms q
40	10	0.079	0.016	-0.006	0.813
40	20	0.116	0.015	-0.017	0.895
40	30	0.118	0.011	-0.008	0.895
40	40	0.114	0.012	-0.007	0.941
40	50	0.106	0.016	-0.015	0.895
50	10	0.095	0.020	-0.015	0.751
50	20	0.123	0.011	-0.015	0.895
50	30	0.161	0.010	-0.013	0.895
50	40	0.134	0.010	-0.004	0.895
50	50	0.132	0.010	-0.017	1.000

Response-window sensitivity of phase prediction

Median phase gains after contextual adjustment were 0.0028, 0.0042, 0.0127, 0.0121, and 0.0152 for 10-, 20-, 30-, 40-, and 50-ms response windows. The corresponding gains after adjustment for measured movement were 0.0023, 0.0028, 0.0073, 0.0077, and 0.0084.

Continuous amplitude-by-phase interaction

This analysis tested whether the cyclic phase profile changed continuously with whisking amplitude. It added interactions between log amplitude and each of the four phase terms after the phase and kinematic main effects. The selected set contained 12,773 touches; 12,296 touches from 35 files and 20 archive subject labels met the model criteria. The median additional fraction of null-model deviance removed was 0.0025 (14/20 labels positive; bootstrap 95% CI, -0.0002 to 0.0150; Wilcoxon $P = 0.030$; sign-test $P = 0.115$). The small median gain and interval spanning zero indicate limited and uneven evidence that the phase profile varied with amplitude.

Supplemental Table S4. Phase-representation sensitivity

Adjustment	Phase representation	Penalty	Median gain	Positive labels	Wilcoxon P	Sign-test P
Context	Two harmonics	0.01	0.0123	13/20	0.154	0.263
Context	Two harmonics	0.1	0.0137	12/20	0.036	0.503
Context	Two harmonics	1	0.0152	15/20	<0.001	0.041
Context	Two harmonics	10	0.0054	15/20	<0.001	0.041
Context	First harmonic	0.1	0.0094	12/20	0.064	0.503
Context	First harmonic	1	0.0097	15/20	0.002	0.041
Context	First harmonic	10	0.0022	15/20	<0.001	0.041
Context	Separate branches	0.1	0.0081	13/20	0.053	0.263
Context	Separate branches	1	0.0055	16/20	<0.001	0.012
Context	Separate branches	10	0.0028	16/20	<0.001	0.012
Measured kinematics	Two harmonics	0.01	0.0070	13/20	0.189	0.263
Measured kinematics	Two harmonics	0.1	0.0137	12/20	0.064	0.503
Measured kinematics	Two harmonics	1	0.0084	15/20	0.002	0.041
Measured kinematics	Two harmonics	10	0.0050	15/20	<0.001	0.041
Measured kinematics	First harmonic	0.1	0.0014	10/20	0.202	1.000
Measured kinematics	First harmonic	1	0.0040	13/20	0.017	0.263
Measured kinematics	First harmonic	10	0.0018	15/20	<0.001	0.041
Measured kinematics	Separate branches	0.1	-0.0007	10/20	0.648	1.000
Measured kinematics	Separate branches	1	0.0027	14/20	0.027	0.115
Measured kinematics	Separate branches	10	0.0020	16/20	<0.001	0.012

The median gain from the reference two-harmonic model was positive at every penalty strength. Directional consistency and signed-rank support varied across settings, so the manuscript treats the phase contribution as small and dependent on model specification.

Supplemental Table S5. Continuous movement and contact coefficients

The table reports archive-label median standardized coefficients from the 50-ms neighboring-onset set. Q values control the false discovery rate across the eight coefficients. Label-level values are supplied in the numerical source data.

Measure	Archive subject labels	Median standardized coefficient	Bootstrap 95% CI	q
Whisking amplitude	20	0.079	-0.012 to 0.107	0.145
Pre-touch angular velocity	20	0.039	-0.003 to 0.107	0.432

Measure	Archive subject labels	Median standardized coefficient	Bootstrap 95% CI	q
Pre-touch angular acceleration	20	0.021	-0.002 to 0.049	0.145
Signed peak curvature change	20	0.038	-0.014 to 0.078	0.440
Absolute peak curvature change	20	0.029	-0.041 to 0.127	0.432
Touch duration	20	0.035	-0.042 to 0.094	0.440
Distance to pole	20	-0.007	-0.043 to 0.025	0.801
Base angle at touch	20	-0.067	-0.179 to 0.138	0.927

Contact duration by phase branch

The reference phase set contained 10,113 protraction touches and 2,660 retraction touches. Contact duration was observed for 10,104 protraction touches and 2,642 retraction touches. Median contact duration was 18 ms during protraction and 44 ms during retraction. Among observed durations, 17.1% of protraction touches and 45.5% of retraction touches lasted at least 50 ms. The neighbor-onset rule removes closely timed new contacts and permits an indexed contact to continue through part of the response window.

Direct branch contrast

The direct retraction-minus-protraction difference in raw spikes per touch was -0.043 (20 archive subject labels; 95% CI, -0.141 to 0.026; $P = 0.202$). The corresponding difference was -0.117 standardized units after contextual adjustment (95% CI, -0.227 to 0.040; $P = 0.143$) and -0.030 after adjustment for measured movement (95% CI, -0.169 to 0.047; $P = 0.245$). These estimates compare weighted branch means within each recording file, followed by one median per archive subject label.

Post-onset contact-description sensitivity

When signed and absolute curvature change and contact duration were added to the predictive adjustment, the median gain from the two-harmonic phase terms was 0.0038 (20 archive subject labels; 12 positive; 95% CI, -0.0023 to 0.0151; $P = 0.070$). The post-onset variables overlap the 50-ms response interval, so this result is reported as a descriptive sensitivity.

Recording-location sensitivity

The reference movement-adjusted model yielded a median phase gain of 0.0210 in 25 C2 files from 12 archive subject labels (95% CI, 0.0084 to 0.0348) and 0.0017 in 10 surrounding-barrel files from 9 labels (95% CI, -0.0035 to 0.0075). One label contributed files to both groups, so the estimates are descriptive subgroup summaries.